

INFO 442 Prototype and Testing Protocol

Group 1

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Prototyping

Our group will create an interactive prototype using Figma before we begin coding.

Acceptance Tests

The acceptance tests our team will perform before beginning user testing goes as follows:

- Feature: Visual Keyboard
 - Process: A user can see the visual keyboard model on their screen and types in letters on their physical keyboard.
 - Pass: The user can see the virtual keyboard and any valid key they select on their physical keyboard, is shown as selected on the virtual keyboard.
 - Fail: The user is unable to see the visual model and when they select keys on their physical keyboard the visual keyboard model doesn't show the user which keys were selected.
- Feature: Auditory Feedback
 - Process: A user types in valid letters to match the word on the screen.
 - Pass: If the user were to get the letter wrong they'll hear a buzz, but if they get it right they'll hear a chime.
 - Fail: The user types an incorrect letter and hears a chime, or they select the correct letter and hear a buzz. If the user selects any key and there is no auditory feedback, we'll know that the acceptance test failed.

Limitations of our acceptance tests goes as follows:

Our testers will likely have corrected or normal vision and therefore we won't be able to test with live color blind or low vision users. Our team is conducting tests on people who are literate in English, which means our in-house testing may indirectly exclude those who struggle with English and users who aren't proficient with the computer. All of our testers, we assume, have full motor function and our acceptance tests may not be able to account for users with physical disabilities.

Users who are deaf, or hard of hearing will not be able to benefit from the Auditory feedback. Since we didn't specify we'd be testing low hearing users, our testers will not be checking for functionality without the auditory feedback. Our testers will be in controlled quiet environments while students may be in loud ones, regardless of hearing abilities. Some computers in labs may be unable to play sound due to technical difficulties.

User Testing

Although our target users are elementary-aged kids with dyslexia, we do not have access to testing our application with children. Therefore, we will strategically conduct user testing by recruiting adult peers who have more experience with early education, or dyslexia. We will aim to get 5-8 users to test our application, and focus on getting as many adults who have or are

familiar with dyslexia, educators or specialists, or general adults if needed. While these users will not entirely replicate children's experience with the app, they can catch confusing instructions, UI, and accessibility issues. We will try to simulate the limitations children may face, asking participants to read at an elementary-aged level, or introducing a time pressure. With our user testing, we will particularly focus on basic usability heuristics, and testing whether the dyslexia-facing features are accessible, such as the text being readable, supportive error feedback, efficiency of the multi-modal aspects, and whether the UI is child-friendly.

Bugs

Bugs that we deem are majorly impacting the usable functionality of our project will be addressed first, then minor bugs will be fixed on by P0, P1, then P2 listings. By utilizing our prioritization list for minor bugs, we ensure that our project meets the minimum criterias we want to focus on the most.

We will re-test the solution after the bug fixes have been completed by ensuring that it passes our unit, integration, and then end-to-end tests, whether they be Jest tests or manual. Additionally, we will ensure that our solution still fulfills all of our P0, P1, and P2 features, such as utilizing a dyslexic-friendly font and a clear and colorful UI. Accessibility and usability will also be re-evaluated to check that the bugs have been fixed for all users.

INFO 442 User Testing Script

Preface: Thank you for meeting with us to do this today! There are no wrong answers, we value any feedback you have for us. We particularly are looking for any frustrations, pain points, and confusions elementary-aged children with dyslexia may have about our first feature.

As you go through the following tasks, please think out loud and walk us through your thoughts, reactions, and expectations as naturally as possible. If possible, please give feedback through the lens of an elementary-aged child with dyslexia.

1. Please view the core typing page. What are your first thoughts as you view the page?
Walk us through any thoughts or questions that pop up.
 - a. *Please note that we are still working on the actual website and UI, and are planning to implement our mascot, accessible/elementary-aged appropriate colors, etc.
 - b. We are planning for the keyboard to be front and center, so the child can easily see it on the screen.
2. Now please try typing in the words displayed on the page. (Let each user type in several words). What are your general thoughts about this feature?

- a. *Note that we are going to implement levels, so this is a general sample of words that may or may not belong to different levels.
- 3. Do you have any thoughts about the font of the words? It is a dyslexic-friendly font, but does the sizing make the words accessible and easy to read?
 - a. *Note that we will be playing around with the color of the font later on.
- 4. IF implemented: Do you have any thoughts about the keyboard itself? Is the UI accessible? What do you think about the functions and features of the keyboard?
 - a.
- 5. Okay, now for some general questions:
 - a. What was easiest to use?
 - b. What was the most difficult?
 - c. What felt missing from the feature?
 - d. Did everything behave as you expected?
 - e. Did everything feel useful or purposeful?
 - f. What would you improve first? What would make this feature more fun or easy to use?
 - g. Would this feel usable for real elementary-aged children with dyslexia?

Conclusion: Thank you all for your feedback, this was very helpful!